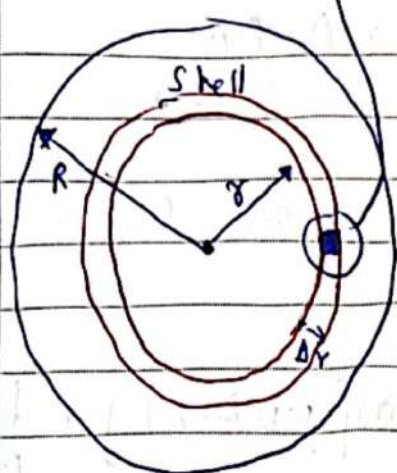


① Gravitational contraction



$$P \Delta A \quad \leftarrow \quad (P + \Delta P) \Delta A$$

two forces
 → self gravity
 → internal pressure

at r → $\rho(r)$ density
 → $P(r)$ pressure

- Radius R , mass M , no rotational motion, Assumed spherical symmetry

$$m(r) = \int_0^r \rho(r') 4\pi r'^2 dr'$$

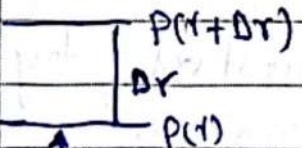
$$g(r) = \frac{GM(r)}{r^2}$$

this is the force ~~of pressure gradient~~ from Newton's law of gravitation, the gravitational force acting on the element of mass ΔM is $g(r) \Delta M$

→ when you go deep pressure changes, pressure at r is $P(r)$ & outer surface $P + \Delta P$

$$\begin{aligned} \text{Net pressure force} &= \left[P(r) + \frac{dP}{dr} \Delta r - P(r) \right] \Delta A \\ &= \frac{dP}{dr} \Delta r \Delta A \end{aligned}$$

$$P = \frac{F}{A}$$



$$F_{\text{inner}} = P(r) \Delta A$$

• pressure decrease when you go out

$$P(r + \Delta r) = P(r) + \frac{dP}{dr} \Delta r$$

mass of the element $DM = \rho(r) DV$
 $= \rho(r) \underbrace{A r DA}_{\text{volume}}$

Net Inward force,

$F = ma \rightarrow DM \frac{d^2 r}{dt^2} = -\text{Grav. force} + \text{Net pressure force}$

$$\rho(r) A r DA \frac{d^2 r}{dt^2} = -g(r) [\rho(r) A r DA] + \frac{dP}{dr} A r DA$$

$$\left(-\frac{d^2 r}{dt^2} \right) + g(r) + \left(\frac{1}{\rho(r)} \frac{dP}{dr} \right)$$

pull of the gravity

net acceleration of the shell matter

Support provided by the pressure gradients.

If a star is in hydrostatic equilibrium (meaning that it is not expanding or contracting) the acceleration is zero

$$\frac{d^2 r}{dt^2} = 0 \rightarrow \text{In this case pressure gradient balances the gravity}$$

$$v = \frac{dr}{dt}$$

$$\frac{dP}{dr} = -\rho(r) g(r)$$

Note: $P(r + \Delta r) \approx P(r) + \frac{dP}{dr} \Delta r$ first order Taylor series expansion

$\rightarrow$ over a very small distance (Δr) the change of

If $p(r)$ differentiable

$$\frac{dp}{dr} = \lim_{\Delta r \rightarrow 0} \frac{P(r + \Delta r) - P(r)}{\Delta r}$$

$$\frac{dp}{dr} \Delta r \approx P(r + \Delta r) - P(r)$$

$$P(r + \Delta r) \approx P(r) + \frac{dp}{dr} \Delta r$$

→ in stellar volume the change of pressure is linear,

$$P(r + \Delta r) = P(r) + \underbrace{\left(\frac{dp}{dr}\right) \Delta r}_{\text{linear term}} + \underbrace{\frac{1}{2!} \left(\frac{d^2 P}{dr^2}\right) \Delta r^2 + \dots}_{\text{curvature of the pressure change}}$$

if $\Delta r \ll$

$\Delta r^2 \rightarrow$ negligible

② Free-fall time (t_{FF})

→ How fast a cloud of gas can collapse under its own gravity

→ speed-limit for star collapse.

→ In reality star collapse $t < t_{FF}$ because of internal pressure

Energy conservation

$$\cancel{\frac{1}{2} m v^2} + \left[\frac{1}{2} \left(\frac{dr^2}{dt^2} \right) \right] = \frac{G m_0}{r} - \frac{G m_0}{r_0}$$

Inward velocity of the shell at radius r , assume mass is constant during collapse.

change of

Note:

Total Energy $E = K + U$

Grav. Potential

for unit mass $= \frac{1}{2} v^2 - \frac{G m_0}{r}$

Energy between a & b

for mass

$$m, \frac{1}{2} m v^2 - \frac{G m m_0}{r}$$

$$t_{ff} = \int_{r_0}^0 \frac{dt}{dr} dr = \int_{r_0}^0 \left[\frac{2Gm_0}{r} - \frac{2Gm_0}{r_0} \right]^{-1/2} dr$$

$$t_{ff} = \sqrt{\frac{8\pi}{32G\rho}}$$

Proof:

$$t_{ff} = \int dt = - \int_{r_0}^0 \left[\frac{2Gm_0}{r} \left(\frac{r_0 - r}{r r_0} \right) \right]^{-1/2} dr$$

$$\text{Let } x = \frac{r}{r_0}, \quad r = r_0 x \quad [dr = r_0 dx]$$

$$\text{B.C. } r = r_0, x = 1$$

$$r = 0, x = 0$$

$$t_{ff} = - \int_1^0 \left[\frac{2Gm_0}{r_0} \left(\frac{1}{x} - 1 \right) \right]^{-1/2} r_0 dx$$

$$t_{ff} = \int_0^1 \left(\frac{r_0}{2Gm_0} \right)^{1/2} \left(\frac{x}{1-x} \right)^{1/2} dx$$

$$t_{ff} = \sqrt{\frac{8\pi}{32G\rho}}$$

→ t_{ff} shows how fast a star forms. If a molecular cloud in space has density ρ , t_{ff} can be obtained w/ the t_{ff} formula

→ If a cloud is transparent it radiates away the heat gravity wins, collapses on the 'free-fall time scale'

→ If a cloud is opaque heat gets trapped, pressure builds up → collapse slows down

↓
"hydrostatic equilibrium"

③ Hydrostatic Equilibrium

[local]

for hydrostatic eqⁿ

for elements of matter

$$\frac{dp}{dr} = - \frac{G m(r) \rho(r)}{r^2} \quad \left| \begin{array}{l} \text{distance} \\ r \end{array} \right.$$

for whole system, this eqⁿ has to be valid at all radii (r)

~~$\int_0^R$~~

$$\int_0^R (\text{something}) \frac{dp}{dr} dr$$

} Integration of pressure gradient term.

$$\Rightarrow \int u dv = \int uv - \int v du$$

$$\text{let } u = 4\pi r^3, \quad du = 12\pi r^2 dr$$

Volume of a thin spherical shell (dv) = $4\pi r^2 dr$

∴ we use $u = 4\pi r^3$ so that math can align to volume weighted average pressure

~~$$\int_0^R p(r) \frac{d}{dr} (4\pi r^3) dr$$~~

$$\frac{dp}{dr} = - \frac{G m(r) \rho(r)}{r^2}$$

$$\int_0^R 4\pi r^3 \frac{dp}{dr} dr = - \int_0^R \frac{G m(r) 4\pi r^2 \rho(r)}{r^2} dr$$

$$E_{GR} = - \int_{m=0}^M \frac{G m(r)}{r} dm$$

use Integration by parts on the left side,

$$[p(r) 4\pi r^3]_0^R - \int_0^R p(r) 12\pi r^2 dr = E_{GR}$$

$p(R) = 0$ at the surface
& $4\pi r^3 = 0$ at the center

$$V = \frac{4\pi R^3}{3}$$

$\langle p \rangle$ volume integrated
avg. pressure

$$-3\langle p \rangle V = E_{GR}$$

$$\langle p \rangle = -\frac{1}{3} \frac{E_{GR}}{V}$$

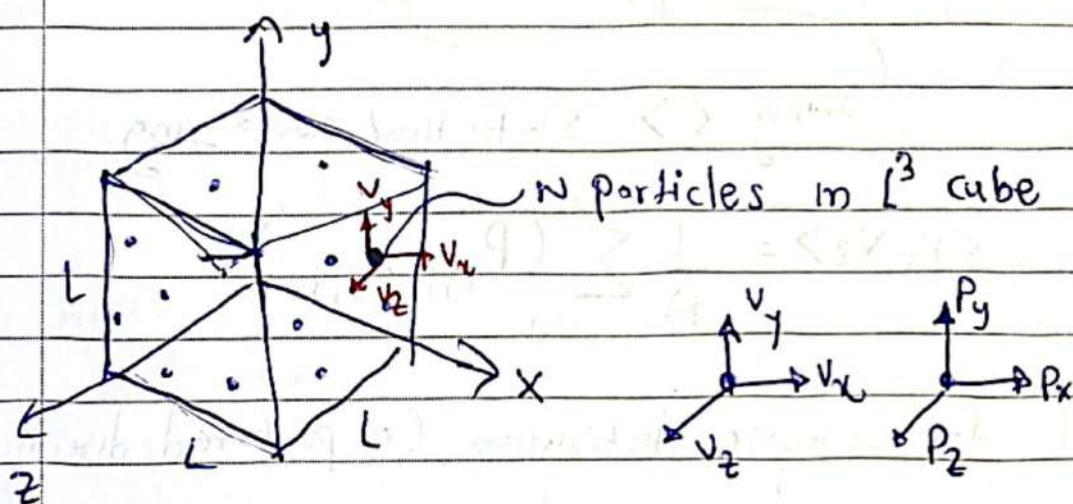
Average pressure is one third of the density of the stored gravitational energy $\rightarrow$ Virial theorem

$\rightarrow$ gravity acts as a "container" & star must generate a specific amount of Internal pressure (P)
balance the weight

How the pressure originated!

Pressure $\rightarrow$ non-relativistic particles $\left\{ \begin{array}{l} \text{classical} \\ \text{quantum} \end{array} \right.$
 $\rightarrow$ ultra-relativistic particles
 $\rightarrow$ translational motion of the particles

① pressure 'P' is generated by particle bouncing off the walls of a container (Stars layers)



elastic collision

$$p_z = m \cdot v_z$$

$\left| \begin{array}{l} \uparrow v_z \\ \downarrow v_z \end{array} \right.$ because the wall is massive & fixed
 particle speed revers

$$\Delta p_z = m v_z - (-m v_z) =$$

$$\Delta p_z = p_z - (-p_z) = 2 p_z$$

time bet^{wn} hits

$$\Delta t = \frac{2L}{v_z} \left\{ \begin{array}{l} \text{average value} \end{array} \right.$$

why distance $2L$)

$$F = \frac{\Delta p}{\Delta t}$$

$$F = \frac{2P_z}{2L/V_z} = \frac{P_z V_z}{L}$$

$$P = F/A = \frac{P_z V_z}{L} \cdot \frac{1}{L^2} = \frac{P_z V_z}{L^3} = \frac{P_z V_z}{V}$$

rate of
momentum
transfer to
unit area

for all N particles in the box,

$$P = \frac{N}{L^3} \langle P_z V_z \rangle$$

why $\langle \rangle$ Statistical averaging

$$\langle P_z V_z \rangle = \frac{1}{N} \sum_{i=1}^N (P_{z,i} V_{z,i})$$

for Isotropic distribution (no preferred direction)

$$\langle P_z V_z \rangle = \langle P_y V_y \rangle = \langle P_x V_x \rangle$$

$$\langle P \cdot V \rangle = \langle P_x V_x + P_y V_y + P_z V_z \rangle$$

$$\langle P_x V_x \rangle = \frac{1}{3} \langle P \cdot V \rangle$$

so pressure on each side of the box is the same

$$n = \frac{N}{V} = \frac{N}{L^3} = \frac{1}{3} n \langle P \cdot V \rangle$$

number of particles
per unit volume

$$P_z = \frac{N}{L^3} \langle p_z v_z \rangle = n \langle p_z v_z \rangle$$

$$= \frac{n}{3} \langle p \cdot v \rangle \leadsto \text{isotropy}$$

Pressure contribution

from the z motion of the particles

= $\frac{1}{3}$ factor act as geometric factor

Pressure in Ideal gas derived using classical physics

↓
This is valid when quantum effects are important
(degenerate e^- gas / relativistic particles)

non/ultra
relativistic
limits

$$\boxed{E_p^2 = m^2 c^4 + p^2 c^2}$$

$$\boxed{v = \frac{p^2 c^2}{E_p}}$$

$$p \ll mc, v = p/m$$

$$E_p = mc^2 + \frac{p^2}{2m} \quad \text{NR (non relativistic)}$$

$$p = mv, p \cdot v = mv^2$$

$$E_k = \frac{1}{2} mv^2$$

$$P = \frac{1}{3} n (mv^2)$$

$$= \frac{2}{3} n \left(\frac{1}{2} mv^2 \right)$$

$$= \frac{2}{3} (\text{Energy density})$$

$$E_p = pc, p \gg mc, v \approx c$$

ultra relativistic (UR)

$$v \approx c, E \approx pc$$

$$p \cdot v = p \cdot c$$

$$P = \frac{1}{3} n pc$$

$$= \frac{1}{3} (\text{Energy density})$$

here energy means translational

- The gas in UR particle exerts less pressure than NR gas for the same amount of internal energy

① Equilibrium of a gas in NR particles

$$\rightarrow \langle P \rangle = \frac{2}{3} \left(\frac{E_{k.E}}{V} \right) \quad \begin{array}{l} \text{translational} \cdot K.E \\ \text{density} \end{array}$$

E.g for hydro. eq^m

$$\rightarrow \langle P \rangle = -\frac{1}{3} \left(\frac{E_{g.R}}{V} \right) \quad \begin{array}{l} \text{gravitational potential} \\ \text{energy density} \end{array}$$

$$\frac{2}{3} \frac{E_{k.E}}{V} = -\frac{1}{3} \frac{E_{g.R}}{V}$$

$$\boxed{2E_{k.E} + E_{g.R} = 0} \quad \text{--- (A)}$$

$$E_{tot} = E_{k.E} + E_{g.R}$$

$$E_{tot} = -E_{k.E}$$

or

$$E_{tot} = \frac{1}{2} E_{g.R}$$

} $\rightarrow \textcircled{B}$

If a star loses the energy by (radiation light), it must contract to satisfy these eq^s

thermostat in stars, If the star radiates energy away and contracts ($E_{g.R}$ become more negative)

$E_{k.E}$ must increase to keep the equation balanced
 -ed $\therefore$ star get hotter as it contracts

For H₅

→ the stability provided by the non-relativistic case is the only reason star can sustain nuclear fusion for millions or billions of years

② Equilibrium of a gas of ultra relativistic particles

→ particles moving near speed of light, like high-energy photons, $e^{\pm}$

$$E_{k \cdot E} + E_{g \cdot R} = 0 \quad \text{--- (A) '}$$

→ the star become (marginally stable), the factor "2" is gone,

→ any slight change (small ~~but~~ losses of energy or change in composition) can cause system to stop being bound and either collapse rapidly or expand uncontrollably

For H.S

(A) Chandrasekar Limit $\leadsto$ type Ia supernova

(B) End of nucleosynthesis

$\hookrightarrow$ collapse

$\hookrightarrow$ Explosive nucleosynthesis

A ③ Equilibrium and the adiabatic Index

$$PV^\gamma = \text{constant}$$

$$dW = -PdV = dE_{\text{IN}} \quad \text{Internal energy}$$

$$E_{\text{IN}} = \frac{1}{1-\gamma} PV$$

from virial theorem

$$\langle P \rangle = -\frac{1}{3} \frac{E_{\text{g.R.}}}{V}$$

$$\langle P \rangle = (\gamma-1) \frac{E_{\text{IN}}}{V}$$

$$\frac{(\gamma-1)E_{\text{IN}}}{V} = -\frac{1}{3} \frac{E_{\text{g.R.}}}{V}$$

$$3(\gamma-1)E_{\text{IN}} + E_{\text{g.R.}} = 0$$

↳ Stability criterion ($\gamma > 4/3$)

$$E_{\text{Tot}} = E_{\text{IN}} + E_{\text{g.R.}}$$

$$E_{\text{Tot}} = (-3\gamma - 4)E_{\text{IN}}$$

$$(3.1) \quad \gamma > 4/3 \rightarrow E_{\text{Tot}} < 0$$

↳ negative total Energy means
system is gravitationally bound

↓
stable self contained
star

(3.2) $\gamma = \frac{4}{3}$, $E_{\text{Tot}} = 0 \rightarrow$ edge case, no binding energy

(3.3) $\gamma < \frac{4}{3}$, $E_{\text{Tot}} > 0 \rightarrow$ unstable, & undergo collapse or expansion

For N.S

standard stars

$$\gamma \approx \frac{5}{3}$$

non relativistic gas are
very stable
long term fusion

$\gamma \rightarrow \frac{4}{3}$: final stage of massive star's life

photo disintegration absorbs massive energy causing

$$\gamma \rightarrow \frac{4}{3}$$